

One scan per film (n = 1, no repeats). Identical protocol throughout: Cu K α λ = 1.54184 Å, 5–50° 2 θ , 0.05° step, 0.3 s per step, Bragg-Brentano geometry, monochromator on.

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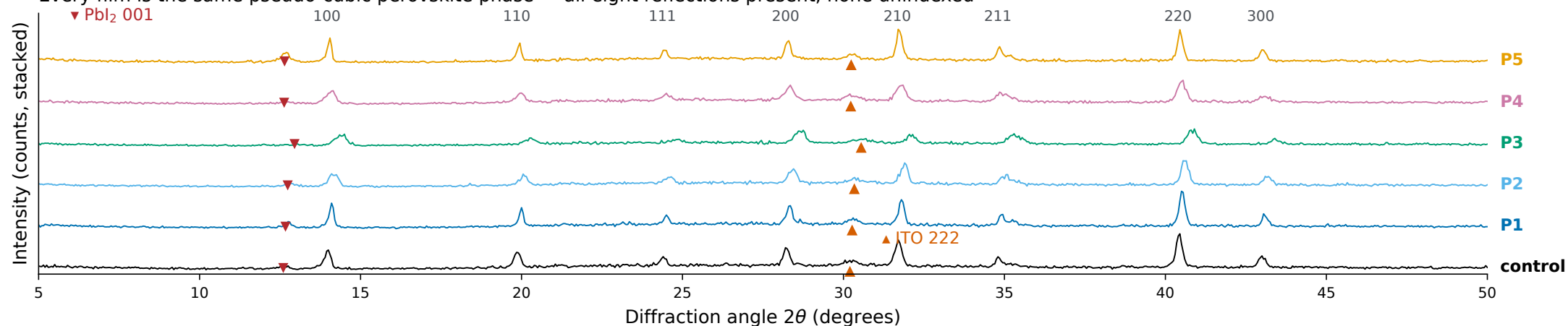
Films measured



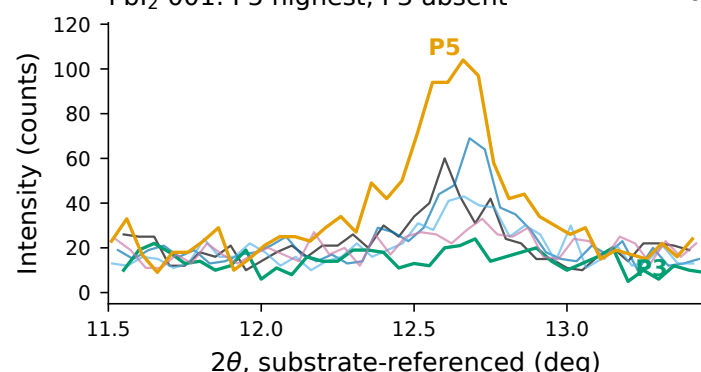
X-ray

The beam samples both layers, so the ITO reflection is an internal reference present in every scan

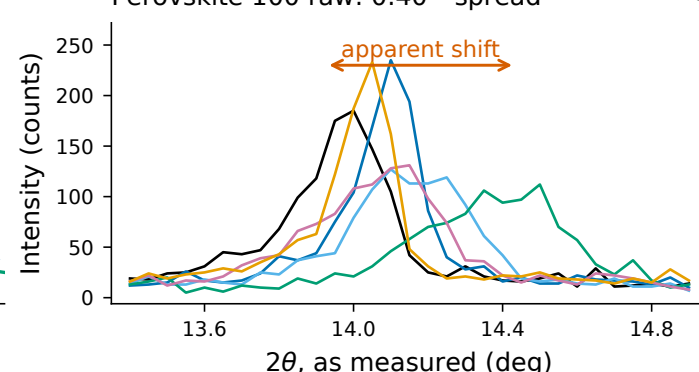
Every film is the same pseudo-cubic perovskite phase — all eight reflections present, none unindexed



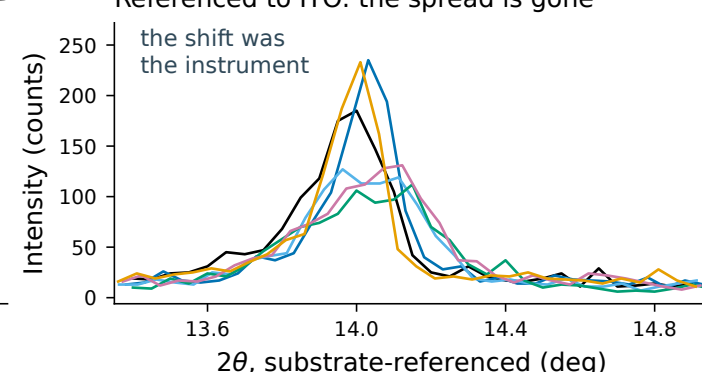
PbI₂ 001: P5 highest. P3 absent



Perovskite 100 raw: 0.40 ° spread

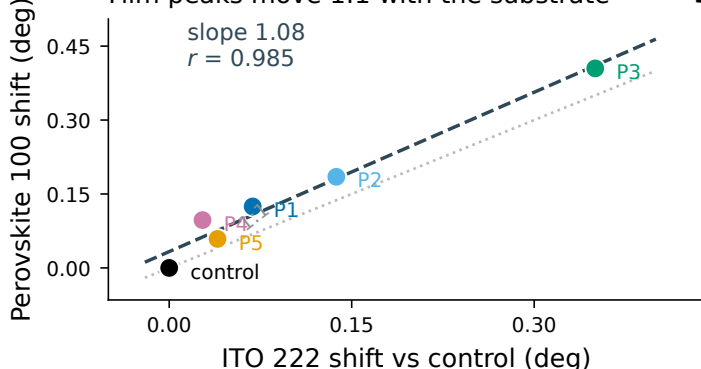


Referenced to ITO: the spread is gone

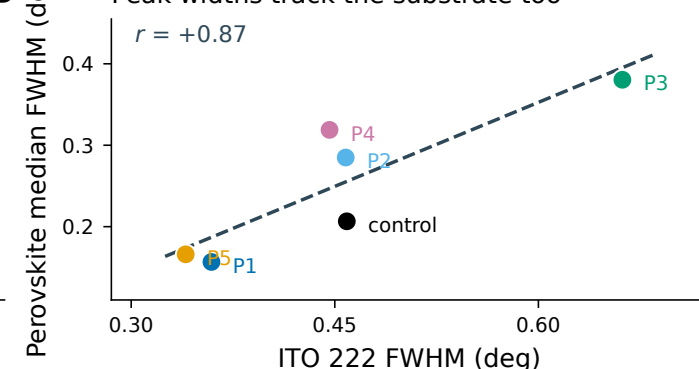


Why the obvious comparisons fail – the substrate cannot change, so any change in its line is instrumental

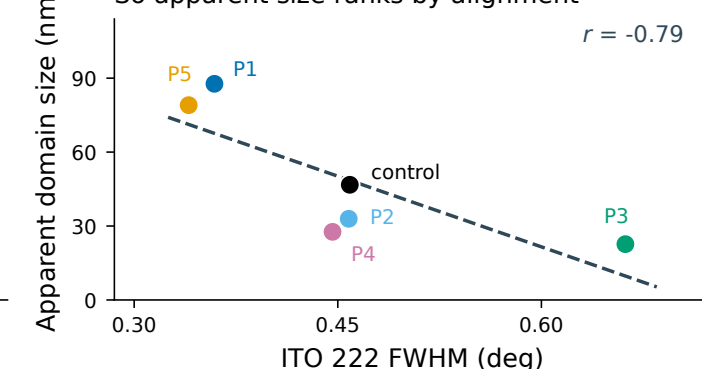
Film peaks move 1:1 with the substrate



Peak widths track the substrate too

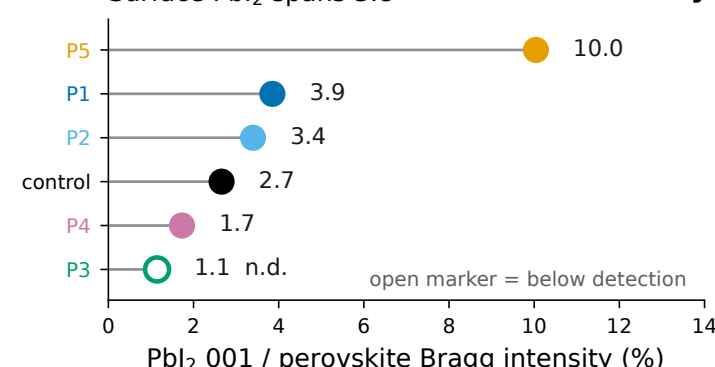


So apparent size ranks by alignment

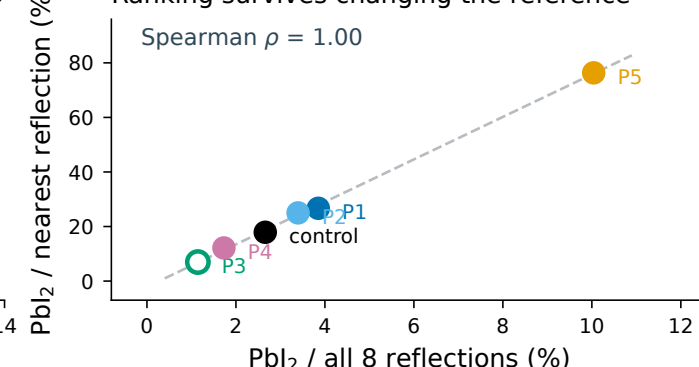


What survives — ratios measured inside a single scan cancel alignment and beam-intensity drift

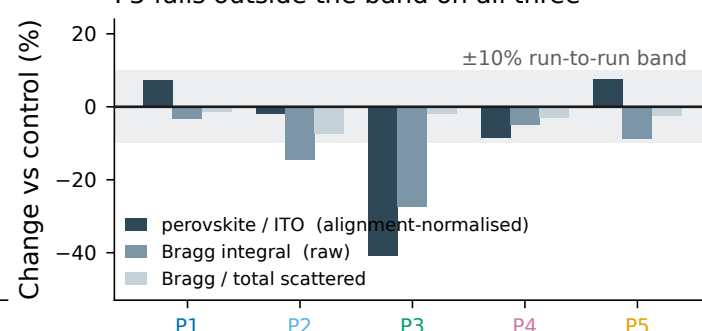
Surface PbI_2 spans $3.8\times$



Ranking survives changing the reference

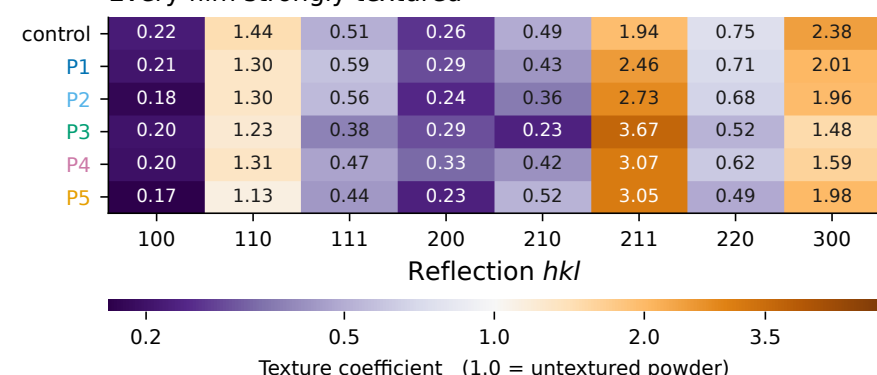
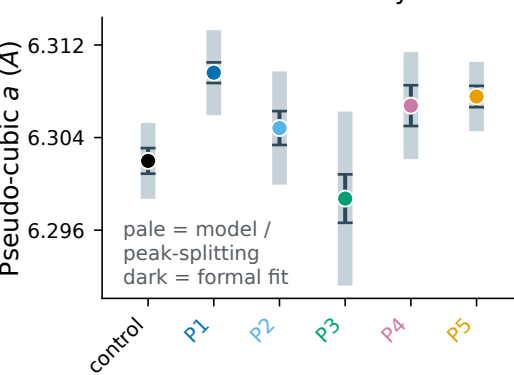


P3 falls outside the band on all three

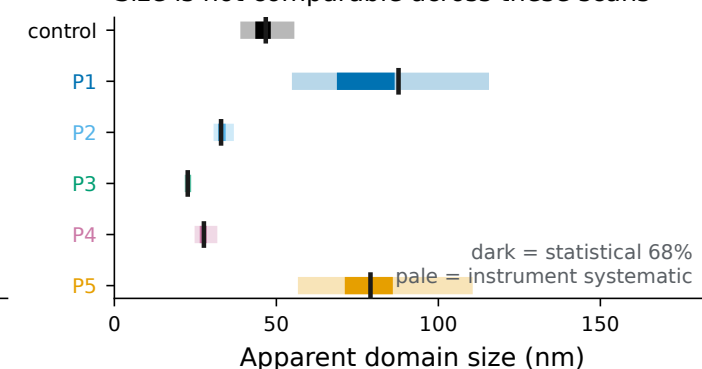


Structure — lattice, texture and size, with each class of uncertainty kept separate

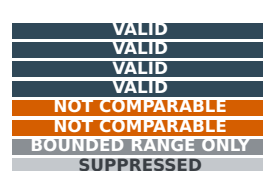
Every film strongly textured

 Δa inside its uncertainty

Size is not comparable across these scans



- Phase identity — same pseudo-cubic cell
- PbI₂ / perovskite Bragg ratio
- Comparative crystallinity indices
- Texture coefficients
- Lattice parameter Δa
- Crystallite size ΔD
- Absolute degree of crystallinity
- Microstrain



all 8 reflections index in all six films
within-scan ratio; ranking robust ($\rho = 1.00$)
protocol identical across all six scans
fixed convention; substrate reflection excluded
peak shifts track the substrate ($1/r = 0.985$)
peak widths track the substrate ($r = +0.87$)
amorphous halo degenerate with background
Williamson-Hall slope negative in 5 of 6